BINUS HIGHER EDUCATION

SMART WATER CHANGE ALERT SYSTEM

Group 1

1. ACHMAD RASYID RAMADHAN
2. ARNAUFALINO IMANDHANI
3. BILLY JOSHUA

SUBJECT MATTER EXPERT

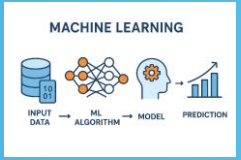
OBJECTIVES & PROJECT SCOPE

Tujuan (*Objectives*):

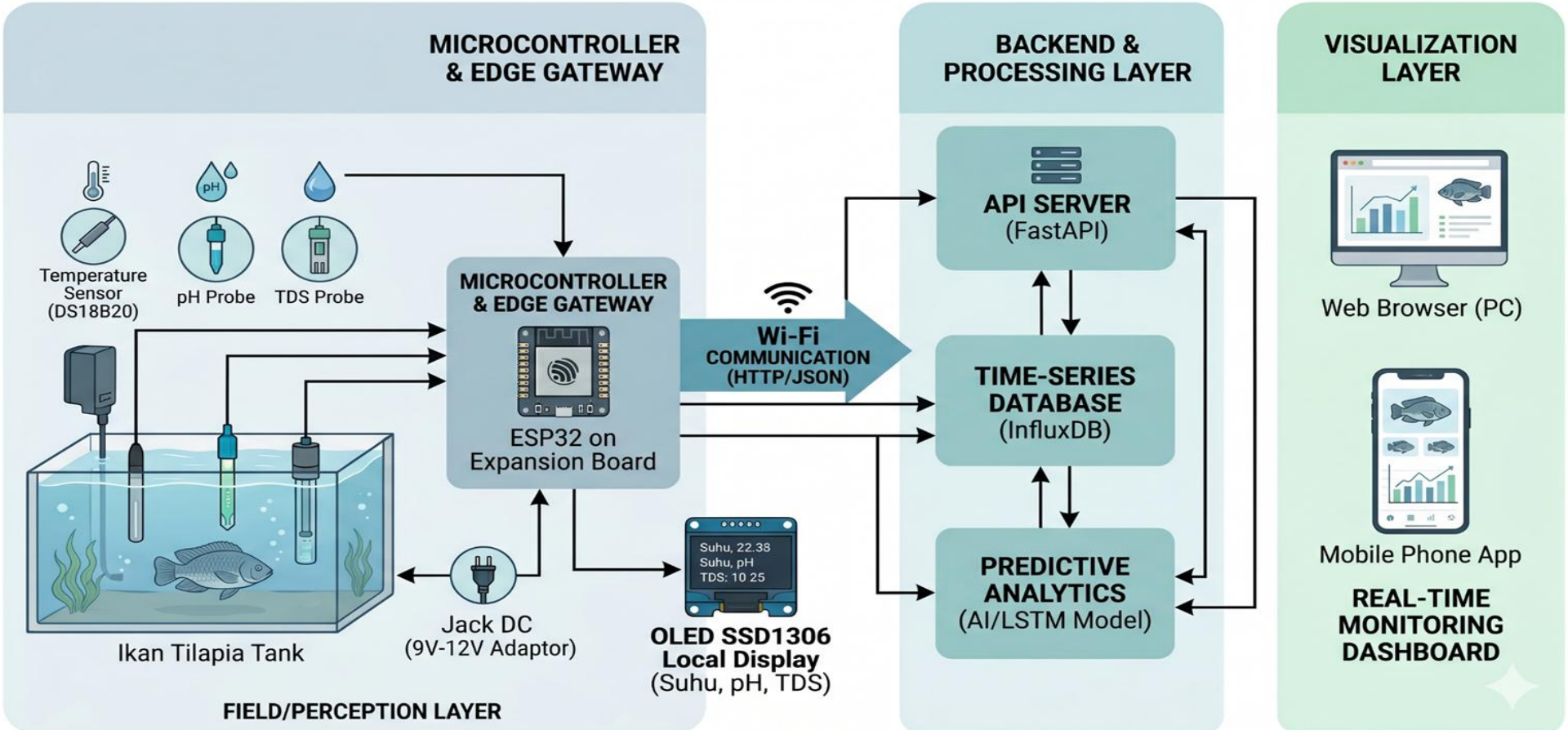
- Merancang dan membangun perangkat IoT (Edge Node) menggunakan ESP32 untuk membaca parameter air secara presisi dan *real-time*.
- Membangun infrastruktur *backend* berbasis FastAPI dan arsitektur *event-driven* (MQTT) untuk ingesti data bervolume tinggi.
- Mengembangkan dan melatih model AI *Deep Learning* (LSTM) untuk memprediksi tren degradasi kualitas air dalam rentang waktu tertentu.
- Mengimplementasikan sistem notifikasi proaktif (*alert system*) yang memberitahu pengguna *sebelum* kualitas air menjadi berbahaya.

Ruang Lingkup (*Project Scope*):

- **Lingkungan:** Pengujian difokuskan pada ekosistem akuarium air tawar berukuran menengah.
- **Parameter Air:** Terbatas pada 4 variabel utama: Tingkat Keasaman (pH), *Total Dissolved Solids* (TDS), Kekeruhan (*Turbidity*), dan Suhu (*Temperature*).
- **Infrastruktur Komputasi:** Pemrosesan *Machine Learning* dan *hosting database* dilakukan pada *Local Server* . Tidak menggunakan layanan *Cloud computing* untuk AI pada fase ini.
- **Fokus Sistem:** Sistem bersifat pemantauan (*monitoring*) dan peringatan (*alerting*). Sistem belum mencakup aktuator otomatis (seperti pompa kuras otomatis) untuk mengganti air.



SYSTEM ARCHITECHTURE DIAGRAM



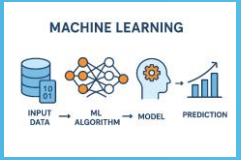
Component List (with specifications)

A. Komponen Perangkat Keras (Hardware)

- **Mikrokontroler:** ESP32 Development Board (Modul WiFi & Bluetooth terintegrasi, ADC 12-bit).
- **Sensor pH:** Modul Sensor pH Analog (misal: PH-4502C) + Probe BNC.
- **Sensor TDS:** Analog TDS Sensor Meter untuk mengukur kepadatan partikel terlarut.
- **Sensor Kekeruhan:** Analog Turbidity Sensor (misal: SEN0189) untuk mengukur tingkat kejernihan air.
- **Sensor Suhu:** DS18B20 Waterproof Temperature Sensor (Komunikasi Digital 1-Wire).
- **Lainnya:** Breadboard, Kabel Jumper (Male-to-Male, Male-to-Female), Resistor 4.7kΩ (Pull-up untuk DS18B20), Power Supply 5V/2A (USB atau Adaptor).

B. Stack Teknologi Perangkat Lunak (Software)

- **Edge Programming:** C++ (Arduino IDE atau PlatformIO).
- **Komunikasi IoT:** Eclipse Mosquitto (MQTT Broker).
- **Backend Framework:** Python 3.10+, FastAPI, Uvicorn, Paho-MQTT.
- **Artificial Intelligence:** TensorFlow / Keras (untuk arsitektur jaringan LSTM), Scikit-learn (untuk *data scaling*), CUDA Toolkit, cuDNN.
- **Database:** InfluxDB atau TimescaleDB (Penyimpanan *Timeseries*), PostgreSQL (Penyimpanan Metadata Model/Konfigurasi), Redis (*In-memory state cache*).
- **Frontend/UI:** Streamlit, Plotly (untuk grafik interaktif).



Procurement Material

Star IoT Store | Chat | Kunjungi Toko | Pesanan tiba di alamat tujuan. diterima oleh Yang bersangkutan. | TELAH DINILAI

Turbidity Sensor Module – Sensor Kekeruhan Air / Kejernihan Air untuk Arduino & Mikrokontroler
x1 — Rp119.900

Total Pesanan: Rp124.376

Menunggu Penjual memberikan nilai | Beli Lagi | Hubungi Penjual | Tampilkan Penilaian Toko

Star+ Supreme_ID | Chat | Kunjungi Toko | Pesanan tiba di alamat tujuan. | TELAH DINILAI

DS18B20 Waterproof Temperature Sensor | 1M cable | 2M Cable | 3M Cable
Variasi: 1 meter
x1 — Rp11.900

ESP32 ESP-32 WROOM WIFI + BLE IoT Development Board (38Pin) CP2102 CH9102X CH340
Variasi: Expansion Board 38p
x1 — Rp29.000

ESP32 ESP-32 WROOM WIFI + BLE IoT Development Board (38Pin) CP2102 CH9102X CH340
Variasi: ESP32 CP2102 USB-C
x1 — Rp82.900

Total Pesanan: Rp128.308

Menunggu Penjual memberikan nilai | Beli Lagi | Ajukan Pengembalian | Lainnya

Belanja 12 Apr 2026 | Selesai | 583498957157532704

Elantech

Resistor Yageo 1k 1/4W 1% 0.25W 1 k 1/4 W 1 Kilo Ohm ElanTech
10 barang x Rp8.700
+7 produk lainnya

Lihat Detail Transaksi | Ulas

Star+ Starlectric | Chat | Kunjungi Toko | Pesanan tiba di alamat tujuan. | TELAH DINILAI

Female To Female Kabel Jumper 10CM
x1 — Rp11.000

Male to Male Kabel Jumper
x1 — Rp15.000

Male to Female Kabel Jumper
x1 — Rp15.000

Star+ Avicena Teknologi Indonesia | Chat | Kunjungi Toko | Pesanan tiba di alamat tujuan. diterima oleh Yang bersangkutan. | TELAH DINILAI

Sensor pH Meter PH-4502C + Probe BNC Elektrode Arduino Raspberry Pi DIY Air Tanah Modul Deteksi Keasaman Air
x1 — Rp221.999

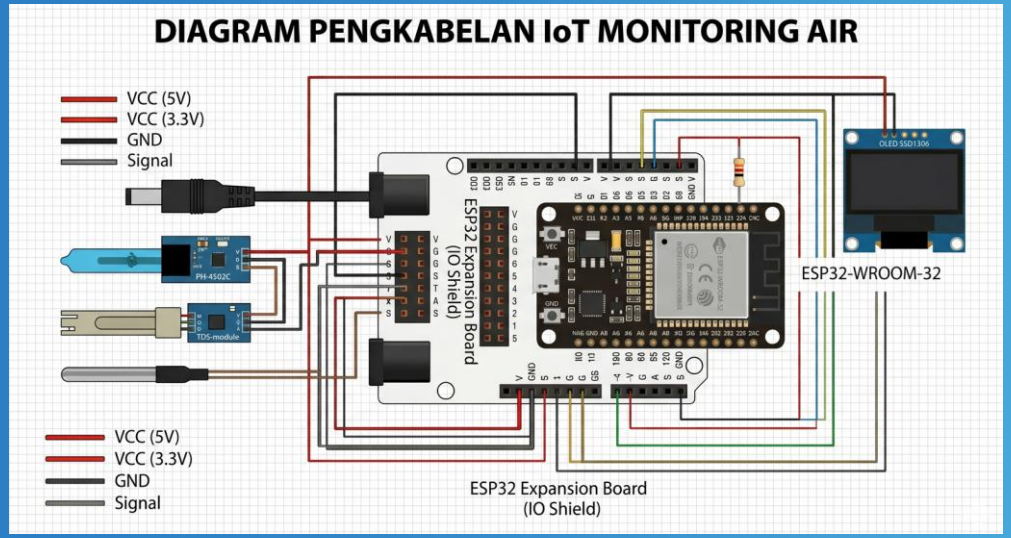
Total Pesanan: Rp227.282

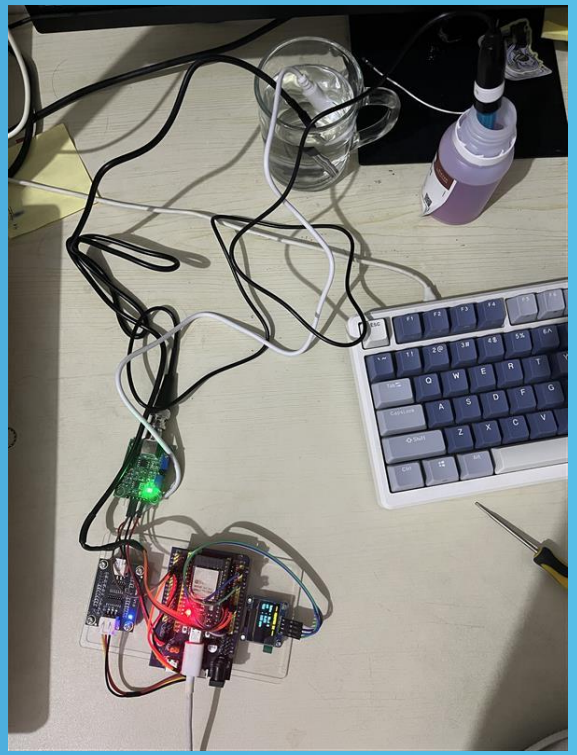
Menunggu Penjual memberikan nilai | Beli Lagi | Ajukan Pengembalian | Lainnya

Pricing part

No	Nama Barang	Variasi/Deskripsi	Qty	Harga Satuan (IDR)	Total Harga (IDR)
1	Serbuk Kalibrasi pH Meter	1 Set (3 Bungkus)	1	15.000	15.000
2	Kabel Jumper Female to Female	10 CM	1	11.000	11.000
3	Kabel Jumper Male to Male	-	1	15.000	15.000
4	Kabel Jumper Male to Female	-	1	15.000	15.000
5	Sensor Suhu DS18B20 Waterproof	Kabel 1 Meter	1	11.900	11.900
6	Expansion Board ESP32	38 Pin	1	29.000	29.000
7	ESP32 WROOM WIFI + BLE	CP2102 USB-C	1	82.900	82.900
8	Sensor pH Meter PH-4502C	+ Probe BNC	1	221.999	221.999
9	TDS Conductivity Meter & Sensor	V1.0 Module	1	79.800	79.800
TOTAL					Rp481.599

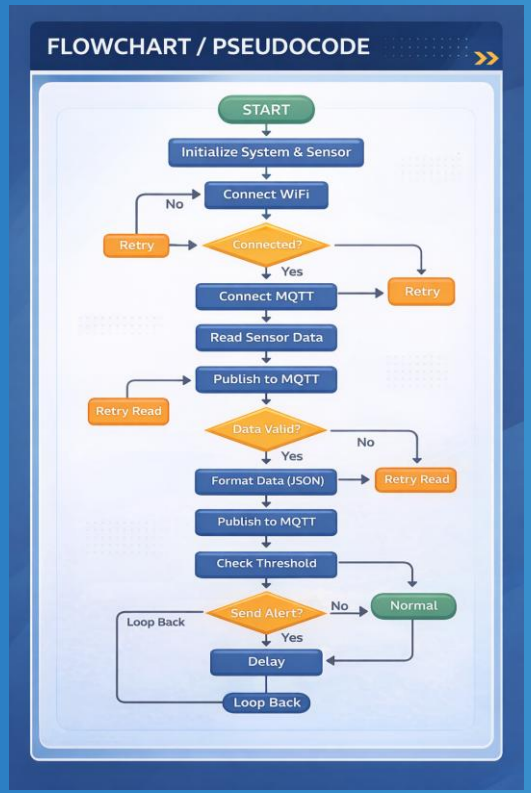
WIRING DIAGRAM



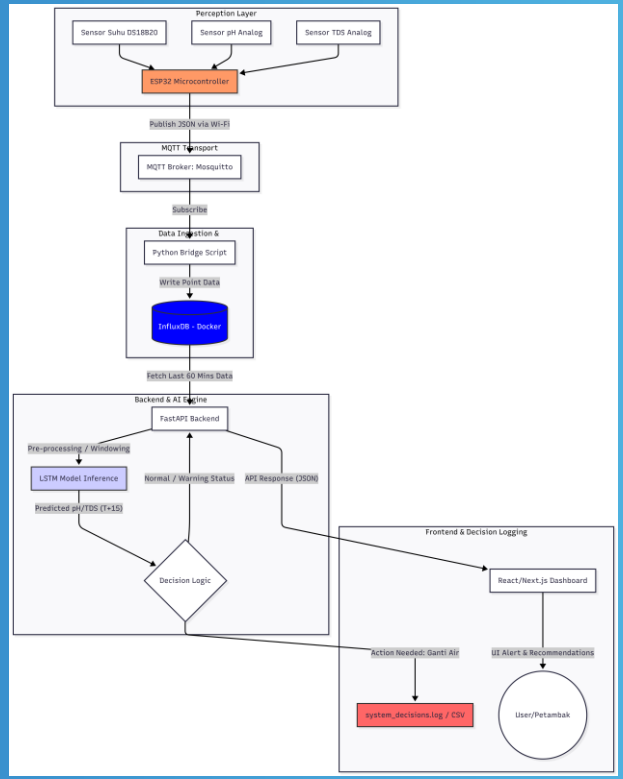


FLOWCHART / PSEUDOCODE

Alur Kerja IOT



Detail-flowchart Hardware & Software



PseudoCode

1. Firmware ESP32 (`sketch_apr13a.ino`)

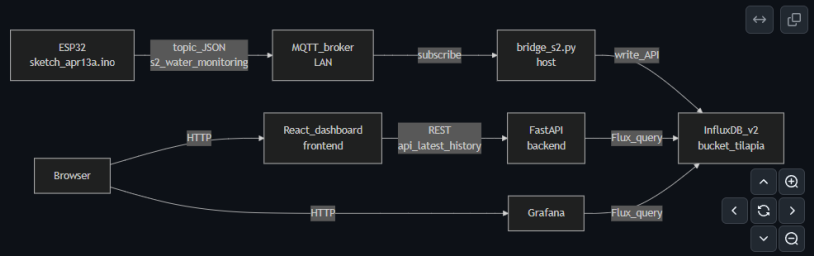
```
PROSEDUR setup():
    inisialisasi Serial, OLED, WiFi
    hubungkan ke WiFi (ssid, password)
    atur server MQTT (alamat, port)
    mulai sensor suhu (OneWire / Dallas)

PROSEDUR loop() — berulang terus:
    JIKA WiFi terhubung MAKA
        JIKA belum terhubung ke broker MQTT MAKA
            reconnect MQTT dengan client_id acak
        akhir JIKA
        jalankan client.loop() untuk menjaga sesi MQTT
    akhir JIKA

    JIKA selisih waktu sekarang dengan kirim_terakhir > 5000 ms MAKA
        tandai kirim_terakhir = sekarang

        baca suhu dari sensor Dallas
        baca TDS dari ADC → voltase → konversi ke ppm (polinomial)
        baca pH dari ADC (rata-rata 10 sampel) → voltase → kalibrasi ke skala pH

        tampilkan suhu, pH, TDS pada OLED

        JIKA MQTT terhubung MAKA
            bentuk string JSON: {"suhu", "ph", "tds"}
            publish(JSON) ke topik "s2/water/monitoring"
        akhir JIKA
    akhir JIKA
```

sketch_apr13a/PSEUDOCODE.md at main · teh05/sketch_apr13a

Flow Software engineering

1. Alur data utama (sensor → penyimpanan → aplikasi)

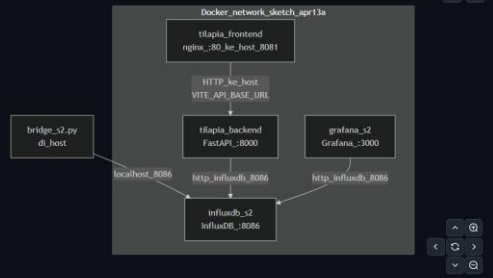
Alur waktu-nyata: sensor di kolam → broker MQTT → skrip bridge menulis ke InfluxDB → API dan dashboard membaca dari InfluxDB.



Payload JSON (contoh): `{"suhu": float, "ph": float, "tds": int}` — selaras dengan firmware dan parser `bridge_s2.py`.

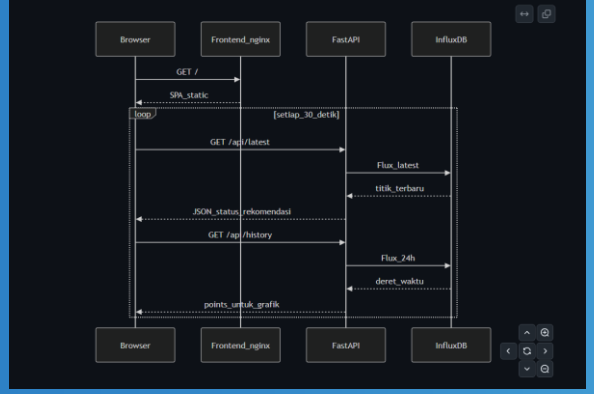
2. Stack Docker Compose (jaringan kontainer)

Layanan yang biasanya dijalankan bersama dengan `docker compose up`. Backend berbicara ke InfluxDB lewat **hostname layanan** `influxdb`, bukan `localhost`.



Layanan	Port host	Peran
`influxdb_s2`	8086	Time-series DB, bucket `tilapia_monitoring`
`grafana_s2`	3000	Dashboard analitis (Flux)
`tilapia_backend`	8000	REST `/api/latest`, `/api/history`, log CSV
`tilapia_frontend`	8081	UI React (static + nginx)

3. Permintaan dashboard (baca data)

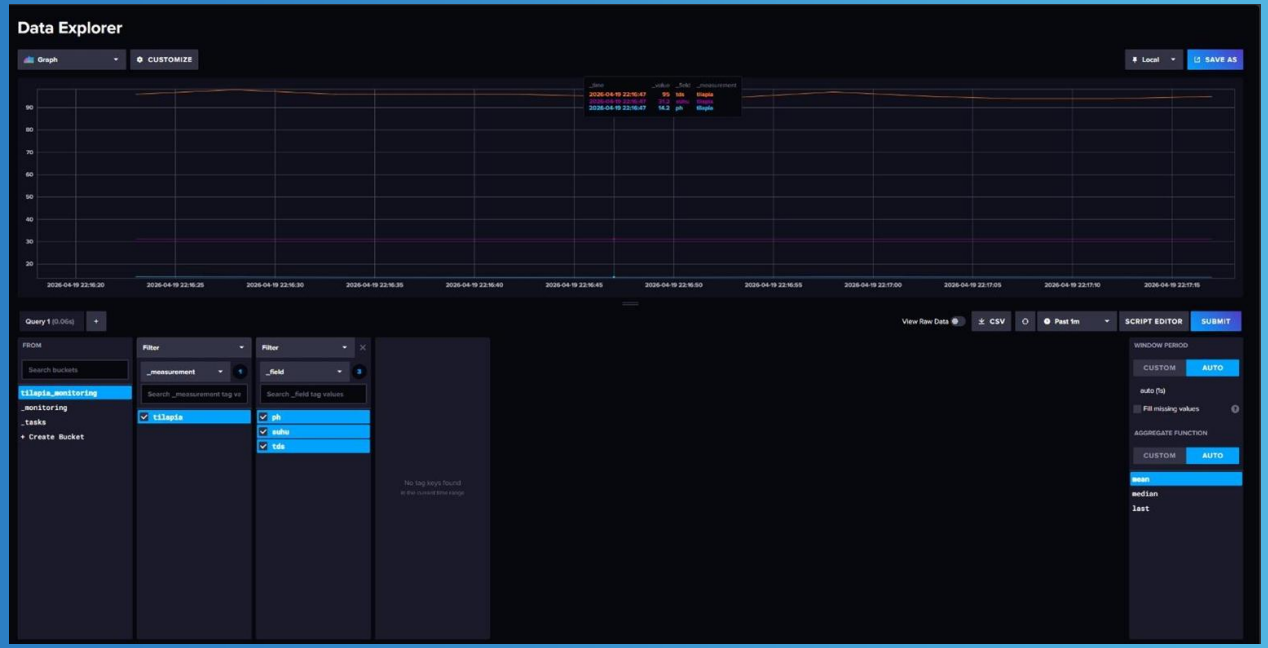


4. File & konfigurasi penting

Lokasi	Fungsi
`sketch_apr13a.ino`	Firmware ESP32, publish MQTT
`bridge_s2.py`	Subscriber MQTT → tulis Influx
`backend/main.py`	API, integrasi `ai_engine`, log `decision_logs.csv`
`frontend/src/App.tsx`	Dashboard, Recharts, polling
`.env` (root)	Token Influx untuk bridge
`backend/.env`	Token & org/bucket untuk API (Docker & lokal)
`docker-compose.yml`	Orkestrasi layanan

SCREENSHOTS OF DASHBOARD

data sudah berhasil masuk di data time series



*masih dalam proses penyempurnaan dan perlu konfigurasi ulang

TESTING PROCEDURE & RESULTS (PROGRESS)

output sensor dari monitor IDE arduino

```
Message (Enter to send message to 'ESP32 Dev Module' on 'COM3')
Suhu: 31.00C | TDS: 67ppm | pH: 3.80 (Volt: 3.115)
Suhu: 31.00C | TDS: 66ppm | pH: 3.53 (Volt: 3.135)
Suhu: 31.00C | TDS: 67ppm | pH: 3.79 (Volt: 3.117)
Suhu: 31.00C | TDS: 67ppm | pH: 3.81 (Volt: 3.115)
Suhu: 31.06C | TDS: 67ppm | pH: 4.15 (Volt: 3.089)
Suhu: 31.06C | TDS: 67ppm | pH: 4.05 (Volt: 3.097)
Suhu: 31.00C | TDS: 68ppm | pH: 3.80 (Volt: 3.116)
Suhu: 31.00C | TDS: 66ppm | pH: 3.72 (Volt: 3.122)
Suhu: 31.00C | TDS: 67ppm | pH: 4.18 (Volt: 3.088)
Suhu: 31.06C | TDS: 66ppm | pH: 4.02 (Volt: 3.100)
Suhu: 31.00C | TDS: 68ppm | pH: 3.78 (Volt: 3.117)
Suhu: 31.00C | TDS: 68ppm | pH: 3.77 (Volt: 3.118)
Suhu: 31.00C | TDS: 66ppm | pH: 4.11 (Volt: 3.093)
Suhu: 31.00C | TDS: 68ppm | pH: 4.14 (Volt: 3.090)
```

TESTING PROCEDURE & RESULTS (PROGRESS)

Connect ke Mosquito

```
a>"C:\Program Files\mosquitto\mosquitto.exe" -v -c "C:\Program Files\mosquitto\mosquitto.
osquitto version 2.1.2 starting
onfig loaded from C:\Program Files\mosquitto\mosquitto.conf.
ridge support available.
ersistence support available.
LS support available.
LS-PSK support available.
ebsockets support available.
pening ipv6 listen socket on port 1883.
pening ipv4 listen socket on port 1883.
osquitto version 2.1.2 running
ew connection from 192.168.43.132:64661 on port 1883.
ew client connected from 192.168.43.132:64661 as ESP32_Aquarium_S2 (p4, c1, k15).
o will message specified.
ending CONNACK to ESP32_Aquarium_S2 (0, 0)
eceived PUBLISH from ESP32_Aquarium_S2 (d0, q0, r0, m0, 's2/water/monitoring', ... (33 by
eceived PUBLISH from ESP32_Aquarium_S2 (d0, q0, r0, m0, 's2/water/monitoring', ... (33 by
eceived PUBLISH from ESP32_Aquarium_S2 (d0, q0, r0, m0, 's2/water/monitoring', ... (33 by
eceived PINGREQ from ESP32_Aquarium_S2
ending PINGRESP to ESP32_Aquarium_S2
eceived PUBLISH from ESP32_Aquarium_S2 (d0, q0, r0, m0, 's2/water/monitoring', ... (33 by
eceived PUBLISH from ESP32_Aquarium_S2 (d0, q0, r0, m0, 's2/water/monitoring', ... (33 by
eceived PUBLISH from ESP32_Aquarium_S2 (d0, q0, r0, m0, 's2/water/monitoring', ... (33 by
```

TESTING PROCEDURE & RESULTS (PROGRESS)

Terdapat problem terhadap daya

```
1776606451: New client connected from 192.168.43.132:59199 as ESP32_Aquarium_S2 (p4, c1, k15).
1776606451: No will message specified.
1776606451: Sending CONNACK to ESP32_Aquarium_S2 (0, 0)
1776606452: Received PUBLISH from ESP32_Aquarium_S2 (d0, q0, r0, m0, 's2/water/monitoring', ... (35 bytes))
1776606457: Received PUBLISH from ESP32_Aquarium_S2 (d0, q0, r0, m0, 's2/water/monitoring', ... (33 bytes))
1776606462: Received PUBLISH from ESP32_Aquarium_S2 (d0, q0, r0, m0, 's2/water/monitoring', ... (33 bytes))
1776606484: Client ESP32_Aquarium_S2 [192.168.43.132:59199] disconnected: exceeded timeout.
```

B. Perbaikan

mengganti sumber daya menggunakan charger Xiaomi 33w 5v

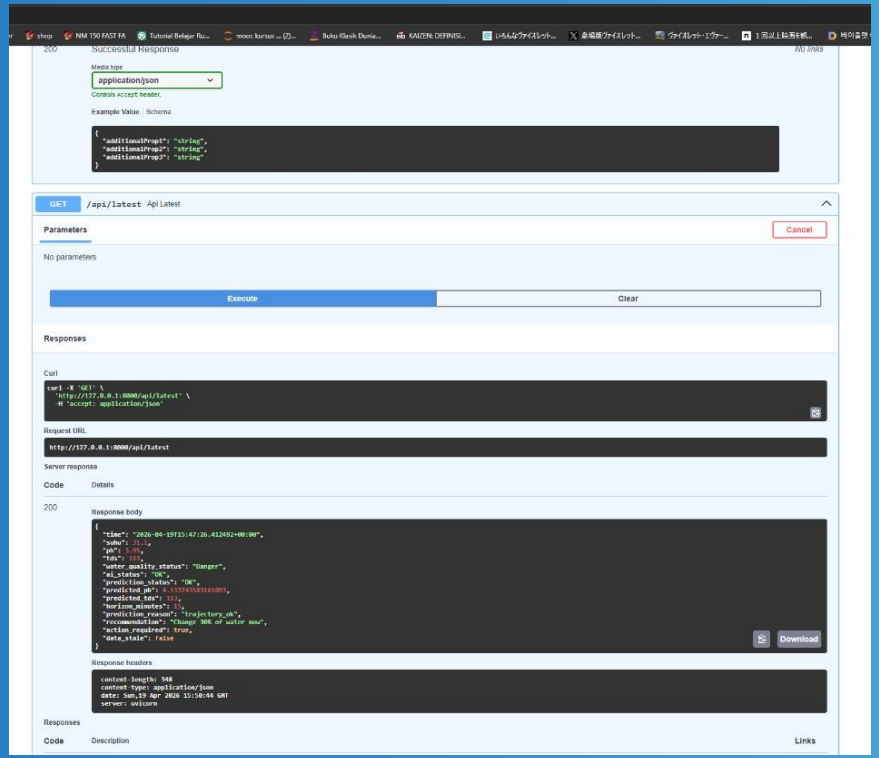
```
s2/water/monitoring {"suhu":31.4,"ph":11.77,"tds":95}
s2/water/monitoring {"suhu":31.3,"ph":11.88,"tds":94}
s2/water/monitoring {"suhu":31.3,"ph":12.05,"tds":95}
s2/water/monitoring {"suhu":31.3,"ph":11.89,"tds":93}
s2/water/monitoring {"suhu":31.3,"ph":11.85,"tds":96}
s2/water/monitoring {"suhu":31.3,"ph":12.21,"tds":96}
s2/water/monitoring {"suhu":31.3,"ph":12.29,"tds":95}
s2/water/monitoring {"suhu":31.3,"ph":12.11,"tds":96}
s2/water/monitoring {"suhu":31.3,"ph":12.10,"tds":98}
s2/water/monitoring {"suhu":31.3,"ph":11.88,"tds":95}
s2/water/monitoring {"suhu":31.3,"ph":12.06,"tds":97}
s2/water/monitoring {"suhu":31.3,"ph":12.23,"tds":94}
s2/water/monitoring {"suhu":31.3,"ph":12.47,"tds":96}
s2/water/monitoring {"suhu":31.3,"ph":12.28,"tds":96}
s2/water/monitoring {"suhu":31.3,"ph":12.46,"tds":97}
s2/water/monitoring {"suhu":31.3,"ph":12.25,"tds":97}
s2/water/monitoring {"suhu":31.3,"ph":12.14,"tds":97}
s2/water/monitoring {"suhu":31.3,"ph":12.55,"tds":98}
s2/water/monitoring {"suhu":31.3,"ph":12.37,"tds":98}
s2/water/monitoring {"suhu":31.3,"ph":12.42,"tds":96}
s2/water/monitoring {"suhu":31.3,"ph":12.56,"tds":98}
s2/water/monitoring {"suhu":31.3,"ph":12.38,"tds":95}
s2/water/monitoring {"suhu":31.3,"ph":12.66,"tds":96}
```

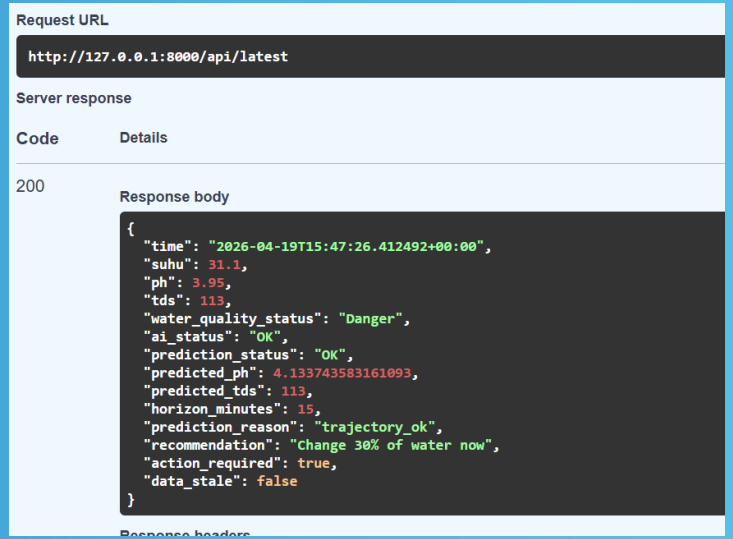
Solved

data parameter sudah dapat terbaca dari sensor

TESTING PROCEDURE & RESULTS (PROGRESS)

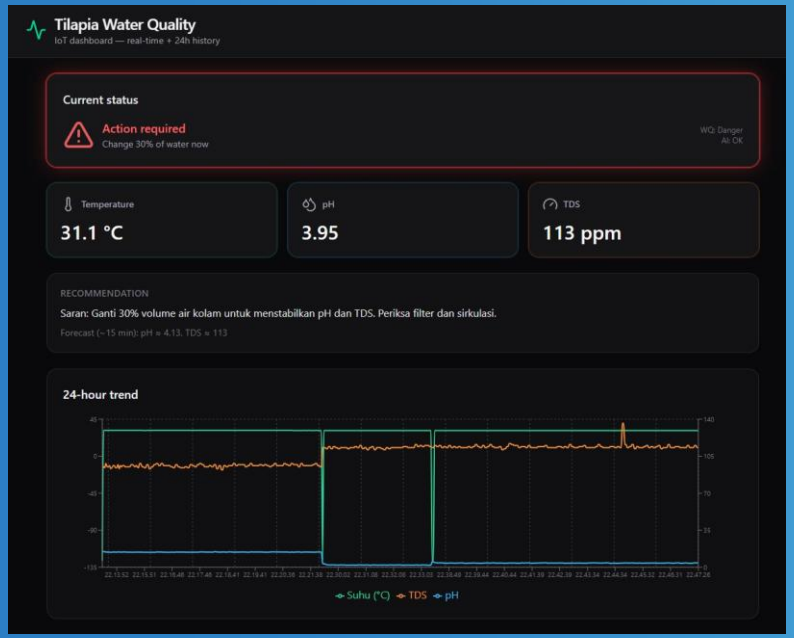
Backend documentation



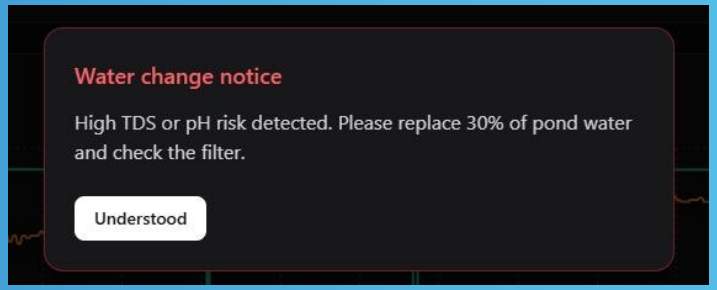


TESTING PROCEDURE & RESULTS (PROGRESS)

Dashboard Frontend



Alert System



CONCLUSION & FUTURE WORK SUGGESTIONS

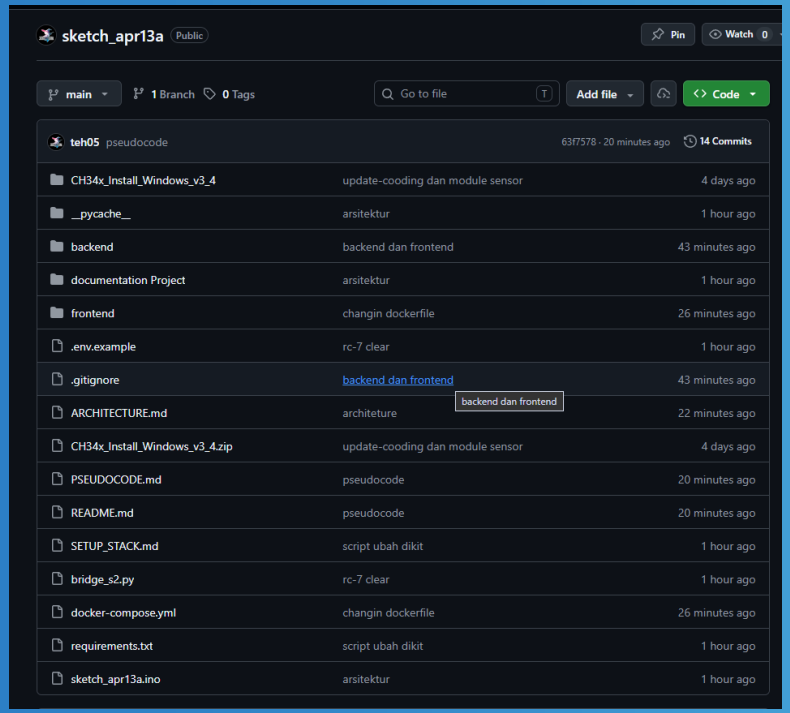
Conclusion

Project ini masih banyak perlu pengembangan Kembali dari sisi optimize logic backend & configuration IOT nya pada PH Salah satu bagian tersulit untuk melakukan optimize project ini

Future Work Suggestions

Pengembangan selanjutnya meliputi integrasi AI, aktuator otomatis, notifikasi lanjutan, dan optimasi sistem.

SOURCE CODE



https://github.com/teh05/sketch_apr13a.git

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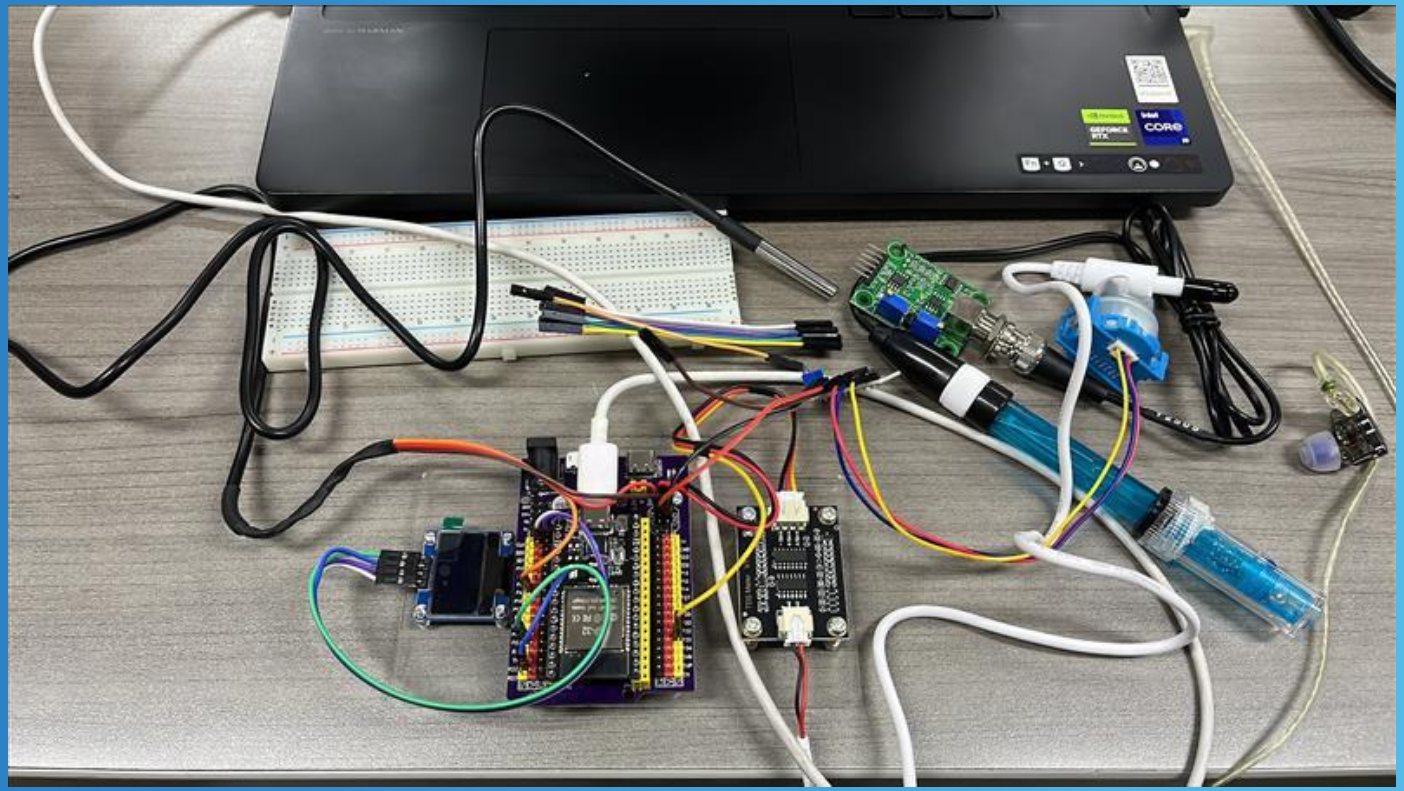
PHOTOGRAPHS